Research — GNOME Platform & libadwaita

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Summary

Build on GNOME 51 (released 2026-09-16) with libadwaita 1.10 and GTK 4.24, in Rust with Blueprint UI files and Meson. libadwaita already provides every desktop piece an Xbox-style library needs (split views, the new `AdwSidebar` from 1.9, breakpoints, toggle groups, status pages), and the cover grid should be a recycling `GtkGridView` over filter and sort list models. GTK has no gamepad input of its own: libmanette (stable 0.2.13; a 1.0 API is in development) is the GNOME answer, and Highscore, Alice Mikhaylenko's unreleased Vala/Blueprint successor to GNOME Games, is the closest reference app.

Two policy facts shape the project: GNOME Circle paused new submissions on 2026-05-29 and adopted an AI policy, and Flathub requires disclosure of AI-generated code and forbids AI-written manifests.

Platform baseline

GNOME 51 is the current stable platform; targeting it gets us `AdwSidebar` (1.9) and everything older.

Component	Version	Date	Notes
GNOME	51 "A Coruña"	2026-09-16	50 on 2026-03-18, 49 on 2025-09-17
libadwaita	1.10.0	September 2026	Main branch docs already say 1.11 alpha
GTK	4.24	about 2026-09-13	GNOME 51 developer notes say "GTK 4.23"; libadwaita 1.10 needs ⩾ 4.23.1; 4.24 in GNOME 51 is inferred
Flathub runtime	`org.gnome.Platform` 51	Autumn 2026	Drops gcr-3 and libhandy

Component	Version	Date	Notes
libmanette	0.2.13 stable; 1.0 alpha on main	2025-10-10	0.2.11 added Steam Deck support via hidraw; 0.2.12 added stick deadzones
Blueprint compiler	0.22	2026	In the GNOME SDK since about GNOME 49; docs still call it experimental
gtk4-rs, libadwaita crate	gtk4 0.11, libadwaita 0.9.2 (features to `v1_10`)	2026-07-07	Needs Rust 1.92+
SDL3	3.4.x	3.4 released 2026-01-05	Alternative gamepad library

What each libadwaita release added

Version	GNOME	Additions relevant to us
1.6	47	`AdwBottomSheet`, `AdwSpinner`, `AdwButtonRow`, `AdwMultiLayoutView`; accent colour API; `.navigation-sidebar` works on `GtkGridView`; default minimum window 360×200
1.7	48	`AdwToggleGroup`, `AdwInlineViewSwitcher`, `AdwWrapBox`; crossfade in view stacks; Adwaita Sans; `.dim-label` renamed `.dimmed`
1.8	49	`AdwShortcutsDialog`; preference groups bound to list models; CSS media queries
1.9	50	`AdwSidebar` (sections, search filter, placeholder, page or sidebar mode), `AdwViewSwitcherSidebar`; view stack sections; reduced-motion support; one `style.css` with media queries replaces dark and high-contrast files
1.10	51	Prefix and suffix slots on `AdwSidebar`; overlay sidebar style class; `AdwCssClassBinding`

Widget toolkit for a game library

Stock widgets cover every need; the one rule that matters for scale is to use `GtkGridView` (recycling) for games and keep `AdwSidebar`, `AdwWrapBox` and boxed lists for small, fixed sets.

Need	Widget(s)	Minimum	Notes
Main window	`AdwApplicationWindow` + `AdwToolbarView`	1.0 / 1.4	`reveal-top-bars` hides bars with animation, useful in fullscreen
Header bar	`AdwHeaderBar`, `AdwWindowTitle`	1.0	Shows back button and page title when a split view collapses
Sidebar plus content	`AdwNavigationSplitView` or `AdwOverlaySplitView` (hideable sidebar)	1.4	Sidebar 25% width within 180–280 sp
Sidebar list	`AdwSidebar` or `AdwViewSwitcherSidebar`	1.9 (slots 1.10)	Built on `GtkListBox`, fine for navigation rows, not for games; switch `mode` to page when collapsed
Library → game details	`AdwNavigationView`, `AdwNavigationPage`	1.4	Push and pop actions, swipe back
A few top-level views	`AdwViewStack` + `AdwViewSwitcher` or `AdwViewSwitcherBar`	1.0	HIG: 3–5 views, otherwise use a sidebar
Filter chips, view toggles	`AdwToggleGroup` (exclusive), `AdwInlineViewSwitcher`; `AdwWrapBox` for chip rows	1.7	`AdwWrapBox` does not recycle, so keep it away from the grid
Cover grid	`GtkGridView` with a signal or builder list item factory	GTK 4.0 (`scroll_to` 4.12)	Builds widgets only for visible items; `max-columns`, `single-click-activate`, `activate` signal

Need	Widget(s)	Minimum	Notes
Model pipeline	`GListStore` → `GtkFilterListModel` → `GtkSortListModel` → selection model	GTK 4.0 (`watch-items` 4.20)	Turn on incremental filtering and sorting for big libraries
Search	`GtkSearchBar` + `GtkSearchEntry` with `key-capture-widget`	GTK 4.0	GTK 4.23 reworked type-to-search
Cover art	`GtkPicture` with `content-fit=cover`	GTK 4.8	Decode untrusted images with glycin
Tile and button styling	`.card`, `.activatable`, `.osd`, `.pill`, `.suggested-action`	1.0+	See style classes
Responsive layout	`AdwBreakpoint`, `AdwMultiLayoutView`	1.4 / 1.6	Set width and height requests yourself; breakpoints remove the minimum size
Dialogs	`AdwDialog`, `AdwAlertDialog`, `AdwPreferencesDialog`, `AdwAboutDialog`	1.5	Become bottom sheets on narrow windows
Feedback	`AdwToastOverlay`, `AdwBanner`, `AdwSpinner`, `AdwStatusPage`	1.0–1.6	Status page is the HIG placeholder
Shortcuts help	`AdwShortcutsDialog`	1.8	Ctrl+?
Theme	`AdwStyleManager` `color-scheme`, `accent-color`	1.6	Prefer-dark is acceptable for media-heavy apps

HIG rules that constrain the design

The HIG pushes us toward a hideable sidebar, a search bar under the header, a few header controls, a list view next to the grid, and a plain name and icon with no trademarks.

- **Sidebars**: use one when there are more views than a switcher holds. The HIG advises against sidebars in rich, immersive content apps, which argues for a hideable sidebar (`AdwOverlaySplitView`) beside a cover-heavy grid.
- **View switchers**: 3–5 views, noun labels in header capitalisation; move to a bottom bar when narrow.
- **Grid views**: a unique thumbnail per item, outlined irregular images, a maximum width, minimal styling, and a list view with metadata as an alternative.
- **Boxed lists**: only for small fixed lists such as settings or game details, never the library.
- **Header bars**: few controls; add and back at the start, title or switcher in the centre, menus at the end; flat icon buttons with tooltips; leave room to drag.
- **Search**: search bar slides down under the header, toggled by a button and Ctrl+F; type-to-search, live and tolerant (fuzzy) matching; "No results" in the content area.
- **Selection mode**: only when three or more actions apply to a selection; Cancel and count in the header, actions in a bottom bar.
- **Placeholders**: an illustrated, upbeat first-run page ("Import games") with a suggested button; plain symbolic placeholders for empty searches and collections.
- **Adaptive**: must work at 1024×600; down to 360×294 if the app should run on phones.
- **Keyboard** and **shortcuts**: everything reachable by keyboard, arrows move in their direction; standard Ctrl+F, Alt+Left, Ctrl+?, Ctrl+, Ctrl+Q, Ctrl+W; avoid Super and Alt combinations. F11 for fullscreen is not on the HIG list.
- **Accessibility**: accessible names everywhere, `alternative-text` on covers; test high contrast, large text, keyboard-only, Orca; never rely on colour alone.
- **Styling**: minimal custom CSS using libadwaita's variables; a brand-green accent would override the user's system accent and goes against the HIG.
- **Naming**: one or two plain nouns under 15 characters; no "GNOME" or "G" prefix, no trademarks (so no "Xbox"), puns, acronyms or invented words.
- **App icons**: a unique metaphor on the 128 px template, no logos, plus symbolic and nightly variants.

Gamepad input and fullscreen in GTK

GTK has no gamepad support, so the app must read controllers itself (libmanette or SDL3) and translate buttons into GTK focus moves and actions, never into synthetic key events.

Why it is hard. Cartridges never shipped controller navigation in a release; its maintainers cited GTK not reading gamepads, Flatpak device access, controllers held exclusively by other programs, and udev permissions (issue #36). Its unreleased rewrite did add libmanette navigation.

libmanette

- Stable 0.2.x (`Manette-0.2`): `ManetteMonitor` with `device-connected` and `device-disconnected` ; `ManetteDevice` with `button-press-event` , `button-release-event` , `absolute-axis-event` , `hat-axis-event` , rumble, device type and user mappings. Mappings come from SDL's controller database.
- 0.2.11 added a native Steam Deck HID driver (turns off "lizard mode", keeps touchpads as pointers); 0.2.12 added stick deadzones.
- 1.0 alpha on main (`Manette-1`) renames signals (`button-pressed` , `absolute-axis-changed`) and adds `list_devices()` ; Highscore builds against it. Release date unknown.
- In Flatpak, libmanette watches `/dev` directly; `--device=input` exposes only `/dev/input` , so the Steam Deck driver still needs `--device=all` . Whether libmanette is in the runtime differs between sources; plan to be able to bundle it. Docs: landing page, 1.0 API, 0.2 Device.

Highscore and prior art

- Highscore lives at gitlab.gnome.org/World/highscore: Vala, Blueprint, libadwaita, libmanette, SDL2, glycin; no Flathub release yet. Alice's posts cover Steam Deck support and touch overlays. Its focus-navigation code could not be read, so its approach is unconfirmed.
- GNOME Games (2017) sent libmanette events from the window to the visible view, which moved a `FlowBox` cursor and activated the child: A and Start activate, D-pad and left stick (threshold ±0.8) move, shoulder buttons switch views, Home goes back. Events had to be queued through the main loop so modal dialogs did not stall them (commit, Games 3.30).

Mapping buttons to GTK

1. D-pad and stick: emit `move-focus` on the window with a `GtkDirectionType` (UP, DOWN, LEFT, RIGHT), the same path arrow keys take (inferred; to verify in a spike).
2. In the grid: `gtk_grid_view_scroll_to(pos, FOCUS | SELECT)` (GTK 4.12) to move and reveal; `list.activate-item` to activate.

3. A: activate the focused widget; B: `navigation.pop` or close the dialog; LB/RB: switch views; Start: menu; Guide: home.
4. Repeat while held (Cartridges' rewrite used a 0.5 deadzone and 280 ms repeat).

Fullscreen and gamescope

- `gtk_window_fullscreen()` is a request the compositor may refuse; watch the `fullscreened` property and add our own `win.fullscreen` action. Combine with `AdwToolbarView:reveal-top-bars = false`.
- Under gamescope, GTK apps run as X11 clients unless gamescope is started with `--expose-wayland`; how libadwaita behaves there is unconfirmed. Useful flags: `-f`, `-W/-H`, `-w/-h`, `-e` (gamescope).

SDL3 alternative: standard gamepad layout, a large mapping database, rumble, LEDs, gyro, touchpads, and `SDL_GetGamepadButtonLabel` for drawing the right A/B or Cross/Circle glyphs (SDL3 gamepad API). Needs the same device access; whether it delivers gamepad events without an SDL window, and how to drive its event pump from the GLib loop, is unconfirmed.

Language and tooling

Recommendation: Rust with gtk4-rs and libadwaita-rs, Blueprint templates, Meson and a GNOME 51 Flatpak, with gamepad input behind one internal interface so libmanette 0.2 can later be swapped for 1.0 or SDL3. The heavy parts (a grid of thousands of covers, background scanning, image decoding, a gamepad loop beside the UI) favour a compiled language with real threads. Vala is the second choice if copying Highscore's patterns matters more than tooling and ecosystem.

Option	Status	Used by	Trade-offs for this app
Rust (gtk-rs book, libadwaita crate, optional Relm4)	Mature; book covers Meson, i18n, D-Bus	Authenticator, Shortwave, Video Trimmer, Resources	Cheap bind and unbind in large grids, real threads plus GLib async; steeper learning curve, slower builds; libmanette 1.0 Rust bindings may need generating with `gir`

Option	Status	Used by	Trade-offs for this app
Vala	Compiles to C, good GIR coverage	Highscore, Calculator, Boxes, Clocks	Near-C speed and Highscore's stack; smaller ecosystem, weaker tooling
Python (PyGObject)	asyncio integration still experimental	Cartridges (archived), Music, Pitivi	Fastest to prototype; costlier bind callbacks on big grids; GIL limits threads
GJS / TypeScript (ts-for-gir)	Supported	Decibels, Weather, Maps	Familiar to web developers; single-threaded; build step for TypeScript
C	Native	Core apps	Most control, highest maintenance cost

Tooling

- Blueprint is in the GNOME SDK, has a language server and editor support, and is used by Highscore and Cartridges.
- Meson is the expected build system; GNOME Builder has "GNOME Application" templates including Rust, updated in GNOME 50.
- Workbench offers Blueprint previews and over 100 demos (including libmanette); Cambalache 1.2 is a visual GTK4 editor.

GNOME Circle and open questions

GNOME Circle is closed to new apps for now, and its criteria are worth meeting anyway because they overlap with Flathub quality.

- **Status**: new submissions paused since 2026-05-29 with no reopening date; Circle also adopted an AI policy where AI may assist but developers must be able to explain their code, and apps that look AI-generated are rejected (committee update).

- **Criteria** (from a third-party copy of the app criteria; the official file could not be opened): core features work without known data loss; actively maintained on a supported runtime; GTK4, libadwaita and the GNOME runtime; follows the HIG with a working dark style; OSI licence, no CLA; valid AppStream and a DOAP file; keyboard navigation, standard shortcuts, high contrast, large text, Orca, tooltips on image buttons; `main` branch, CI, public issue tracker, Code of Conduct; on Flathub for x86_64 and aarch64; no GNOME branding; translations via a localisation platform.
- **Fit**: a launcher like this is plausible (Cartridges was in Circle), but Xbox-style naming would fail the trademark and naming rules, and `--device=all` may draw scrutiny.

Open questions

- Exact GTK point release in GNOME 51.
- Highscore's gamepad navigation approach, Flatpak permissions and release plans.
- libmanette 1.0 release date, and whether it will include UI helpers.
- Whether emitting `move-focus` gives reliable spatial navigation inside `GtkGridView`.
- How libadwaita apps behave under gamescope.
- Whether SDL3 delivers gamepad events without its own window inside a GTK app.

Sources

- Releases: Libadwaita 1.6, 1.7, 1.8, 1.9; GNOME 51, GNOME 51 developers, GNOME 50 developers, GNOME 51 released, GNOME 51 runtime changes; Arch libadwaita 1.10.0; GTK 4.22, GTK 4.24, Scalable lists in GTK 4
- libadwaita docs: index, Sidebar, ViewSwitcherSidebar, NavigationSplitView, Breakpoint, ToolbarView, WrapBox, ToggleGroup, ShortcutsDialog, style classes, styles and appearance, adaptive layouts
- GTK docs: GridView, ListView, FilterListModel, SortListModel, SearchEntry, Picture, move-focus, DirectionType, ListScrollFlags, Window.fullscreen
- HIG: Sidebars, View switchers, Grid views, Boxed lists, Header bars, Search, Selection mode, Placeholders, Windows, Adaptive, Keyboard, Standard shortcuts, UI styling, Accessibility, App naming, App icons
- libmanette: landing page, 1.0 docs, Device, Monitor, 0.2 Device (PyGObject), GitHub mirror, Workbench demo
- Highscore and GNOME Games: Steam Deck, HID and libmanette, mobile testing in libadwaita, nixpkgs package, highscore-emu, gamepad navigation commit (2017), Games 3.30, GamepadNavigation design wiki

- Cartridges: issue #36 (gamepad), archived repo
- Discussion: couch and handheld gaming on GNOME, gamepad TV mode, Flathub device=input, Flatpak 1.16
- gamescope and SDL: gamescope, gamescope wiki, Mozilla bug 1802802, SDL3 gamepad API, SDL 3.4
- Languages and tooling: GNOME languages, gtk-rs book, libadwaita crate features, Relm4, PyGObject async, ts-for-gir, Blueprint, Workbench
- Circle and policy: circle.gnome.org, Circle committee update, Circle handbook, criteria checklist, Flathub requirements, Flathub reverses AI ban